Name: _____

WORKSHEET

Antiderivatives

1 Given the following gradient functions, find the antiderivatives.

a $f'(x) = 3$ _____

b $f'(x) = 6x$ _____

c $f'(x) = -2 - 5x$ _____

d $f'(x) = 6x^2$ _____

e $f'(x) = -2x^2 + 5 + 9x$ _____

f $f'(x) = -24x^3$ _____

g $f'(x) = x^3 - 7x^2 + 8x - 1$ _____

h $f'(x) = 5x^4 - 6x^3 + x - 12$ _____

i $f'(x) = x^5 + \pi x^3 - 7x$ _____

j $f'(x) = \sqrt{2}\, x^4 + 16x^3 - 10x^2$ _____

2 Find the antiderivatives of the following

a $y(x) = 3x^2 - 5x + 2$ _____

b $f(t) = -3t^2 + 5t - 7$ _____

c $p(i) = 4i^3 - 8i + 3$ _____

d $z(j) = j^9 - 7j^6 + j^2 - 3$ _____

e $a(r) = \frac{1}{2}r^3 - \frac{2}{3}r^2 + \frac{4}{7}r + \frac{5}{9}$ _____

f $q(t) = 8t^{10} - 6\sqrt{3}\, t^5 - 4t^2 + 1$ _____

g $f(t) = t^5 + 3t^3 - 4t - 2$ _____

h $g(k) = 9k^5 + 15k^4 - 10k^3 + 12k^2 + 3$ _____

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3 Find the antiderivatives of the following.

a $f(x) = (x - 5)(x + 2)$

b $f(x) = (x^2 - 2x)(3 - x)$

c $f(x) = (1 - x^3)(x^2 + 4)$

d $f(x) = (3x^2 + 7x)(2x - x^2)$

e $f(x) = (x + 4)^2$

f $f(x) = (5 - x^2)^2$

g $f(x) = (2x^2 - 3x)^2(x + 1)$

h $f(x) = (x^2 + 3x)^3(2x - 5)$

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Answers

1 **a** $f(x) = 3x + c$

b $f(x) = 3x^2 + c$

c $f(x) = -\frac{5}{2}x^2 - 2x + c$

d $f(x) = x^3 + c$

e $f(x) = -\frac{2}{3}x^3 + \frac{9}{2}x^2 + 5x + c$

f $f(x) = 6x^4 + c$

g $f(x) = \frac{x^4}{4} - \frac{7x^3}{3} + 4x^2 - x + c$

h $f(x) = x^5 - \frac{3}{2}x^4 + \frac{x^2}{2} - 12x + c$

i $f(x) = \frac{x^6}{6} + \frac{\pi x^4}{4} - \frac{7x^2}{2} + c$

j $f(x) = \frac{\sqrt{2}x^5}{5} + 4x^4 - \frac{10x^3}{3} + c$

2 **a** $Y(x) = x^3 - \frac{5x^2}{2} + 2x + c$

b $F(t) = -\frac{3t^4}{4} + \frac{5t^2}{2} - 7t + c$

c $P(i) = i^4 - 4i^2 + 3i + c$

d $Z(j) = \frac{j^{10}}{10} - j^7 + \frac{j^3}{3} - 3j + c$

e $A(r) = \frac{r^4}{8} - \frac{2r^3}{9} + \frac{2r^2}{7} + \frac{5r}{9} + c$

f $Q(t) = \frac{8t^{11}}{11} - \sqrt{3}t^6 - \frac{4t^3}{3} + t + c$

g $F(t) = \frac{t^6}{6} + \frac{3t^4}{4} - 2t^2 - 2t + c$

h $G(k) = \frac{3k^6}{2} + 3k^5 - \frac{5k^4}{2} + 4k^3 - \frac{5k^2}{2} + 3k + c$

3 **a** $F(x) = \frac{x^3}{3} - \frac{3x^2}{2} - 10x + c$

b $F(x) = -\frac{x^4}{4} + \frac{5x^3}{3} - 3x^2 + c$

c $F(x) = -\frac{x^6}{6} - x^4 + \frac{x^3}{3} + 4x + c$

d $F(x) = -\frac{3x^5}{5} - \frac{x^4}{4} + \frac{14x^3}{3} + c$

e $F(x) = \frac{x^3}{3} + 4x^2 + 16x + c$

f $F(x) = \frac{x^5}{5} - \frac{10x^3}{3} + 25x + c$

g $F(x) = \frac{2x^6}{3} - \frac{8x^5}{5} - \frac{3x^4}{4} + 3x^3 + c$

h $F(x) = \frac{x^8}{4} + \frac{13x^7}{7} + \frac{3x^6}{2} - \frac{81x^5}{5} - \frac{135x^4}{4} + c$

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